Warnaar's cylindric conjecture for the balanced profile $(4, 3, 3)$ at modulus 13: a complete proof via Warnaar's finite form ($a = +1$) and Uncu's modulo-13 theorem

Seed 1, Layer 4, Round 2 (loop experiment 2026-07-04)

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Abstract

We prove the case $k = 4$ of Warnaar's Conjecture `Con_cylindric-b` (the A_2 Andrews–Gordon fermionic-form conjecture for cylindric partitions of balanced profile $(k, k-1, k-1)$ at modulus $3k+1$): the profile $c = (4, 3, 3)$ at level $d = 10$ (modulus 13). The proof chains two proved results: Warnaar's finite-form Proposition ($a = +1$, $k = 3$) in a coefficientwise limit, and Uncu's 2024 modulo-13 theorem expressing $H_{(4,3,3)}$ in the series $S_{13}(\rho|\sigma)$. Unlike the $d = 8$ balanced case, *no* Kanade–Russell contiguous relation is needed: after the Pochhammer split, the limit of the finite form coincides verbatim with Uncu's proved expression. As a corollary the coefficients $Q_{n,(4,3,3)}(q)$ are manifestly nonnegative: the second proved balanced case of the fermionic-form conjecture, the first proved case of `Con_cylindric-b` beyond $k = 2$, and the first proved-positive core orbit at $d = 10$.

1 Setup and statement

Throughout, $(q;q)_n = \prod_{i=1}^{n}(1-q^i)$, and $\begin{bmatrix} n \\ m \end{bmatrix}$ is the q-binomial coefficient (zero unless $0 \le m \le n$). For a profile $c = (c_1, c_2, c_3)$ with $c_1 + c_2 + c_3 = d$, let $F_c(z, q)$ be the Corteel–Welsh cylindric-partition generating function (raw interlacing definition; z tracks the largest part, q the size), and set

$$G_c(z,q) = (zq;q)_\infty F_c(z,q), \qquad H_c(z,q) = \frac{(zq;q)_\infty}{(q;q)_\infty} F_c(z,q) = \frac{G_c(z,q)}{(q;q)_\infty},$$

$$Q_{n,c}(q) = (q;q)_n \, [z^n] \, G_c(z,q).$$

Warnaar's positivity conjecture (Corteel–Dousse–Uncu) asserts $Q_{n,c}(q) \ge 0$ coefficientwise for all n, c with $d \not\equiv 0 \pmod 3$. For the balanced profile $(k, k-1, k-1)$ of modulus $3k+1$, Warnaar's Conjecture `Con_cylindric-b` (first identity) predicts the fermionic form

$$G_{(k,k-1,k-1)}(z,q) \overset{?}{=} \mathrm{FERM}^+_{k-1}(z,q) := \sum_{\substack{n_1,\ldots,n_{k-1}\ge 0 \\ m_1,\ldots,m_{k-1}\ge 0}} \frac{z^{n_1} q^{\sum_{i=1}^{k-1}(n_i^2 - n_i m_i + m_i^2)}}{(q;q)_{n_1}} \begin{bmatrix} 2n_{k-1} \\ m_{k-1} \end{bmatrix} \prod_{i=1}^{k-2} \begin{bmatrix} n_i \\ n_{i+1} \end{bmatrix} \begin{bmatrix} n_i - n_{i+1} + m_{i+1} \\ m_i \end{bmatrix} \tag{1}$$

The case $k = 1$ is trivial and $k = 2$ is Corteel–Welsh [1, Theorem 3.2]; all $k \ge 3$ were open. We prove the case $k = 4$, profile $(4, 3, 3)$, $d = 10$, modulus 13.

For $m = 13$ ($k = 4$ in Uncu's indexing $m = 3k+1$), $\rho, \sigma \in \mathbb{Z}^3$, define (Uncu 2024, eq. (`eq:Sp1`); note there is *no* $q^{2r_3s_3}$ cross term, in contrast to S_{3k-1})

$$S_{13}(z;\rho|\sigma) = \sum_{\substack{r_1\geq r_2\geq r_3\geq 0\\ s_1\geq s_2\geq s_3\geq 0}} z^{r_1} \frac{q^{\sum_{i=1}^{3}(r_i^2 - r_is_i + s_i^2 + \rho_ir_i + \sigma_is_i)}}{(q;q)_{r_1-r_2}(q;q)_{r_2-r_3}(q;q)_{r_3}\,(q;q)_{s_1-s_2}(q;q)_{s_2-s_3}(q;q)_{s_3}\,(q;q)_{r_3+s_3+1}}, \tag{2}$$

and $e_3 = (0,0,0)$, $e_2 = (0,0,1)$. We suppress z and write $S(\rho|\sigma)$.

Theorem 1. *$G_{(4,3,3)}(z,q) = \mathrm{FERM}_3^+(z,q)$, i.e. Warnaar's Conjecture* `Con_cylindric-b` *(first identity) holds for $k = 4$.*

Corollary 2. *For all $n \geq 0$,*

$$Q_{n,(4,3,3)}(q) = \sum_{\substack{n_2,n_3\geq 0\\ m_1,m_2,m_3\geq 0}} q^{n^2+n_2^2+n_3^2-nm_1-n_2m_2-n_3m_3+m_1^2+m_2^2+m_3^2} \begin{bmatrix} n\\ n_2\end{bmatrix}\begin{bmatrix} n-n_2+m_2\\ m_1\end{bmatrix}\begin{bmatrix} n_2\\ n_3\end{bmatrix}\begin{bmatrix} n_2-n_3+m_3\\ m_2\end{bmatrix}\begin{bmatrix} 2n_3\\ m_3\end{bmatrix},$$

a manifestly nonnegative polynomial (each $n_i^2 - n_im_i + m_i^2 \geq 0$, all q-binomials nonnegative). In particular Warnaar's positivity conjecture holds for the profile $(4,3,3)$ at $d = 10$.

Proof of the corollary. $Q_{n,(4,3,3)} = (q;q)_n[z^n]\mathrm{FERM}_3^+$; the summand of (1) at $n_1 = n$ has denominator $(q;q)_{n_1} = (q;q)_n$, which cancels. □

Remark 3. *The profile $(4,3,3)$ is chirality-safe: its reversal $(3,3,4)$ is a cyclic shift of $(4,3,3)$, and F_c is invariant under cyclic shifts of the profile. No orbit-labelling subtleties (synthesis-layer3 §4(iv)) arise.*

2 Proof of Theorem 1

The proof chains three proved statements. In contrast to the $d = 8$ balanced case (Seed 2, Layer 3), no bridging contiguous relation is required: the modulus-13 family has no $q^{2r_3s_3}$ cross term, and the Pochhammer split lands directly on Uncu's proved expression.

Step 1: Warnaar's finite form ($a = +1$) in the limit

Warnaar (A_2 Andrews–Gordon paper, Section 7) defines, for $\sigma = (\sigma_1,\ldots,\sigma_k) = (1,\ldots,1,a)$, $a \in \{-1,1\}$,

$$F^{(a)}_{n_0,m_0;k}(z,q) = \sum_{\substack{n_1,\ldots,n_k\geq 0\\ m_1,\ldots,m_k\geq 0}} \frac{z^{n_1}}{(q;q)_{n_k+m_k}} \prod_{i=1}^{k} q^{n_i^2-\sigma_in_im_i+m_i^2}\begin{bmatrix} n_{i-1}\\ n_i\end{bmatrix}\begin{bmatrix} m_{i-1}\\ m_i\end{bmatrix},$$

and proves (his Proposition `Prop_finiteform`, case $a = +1$, in which $\sigma_i = 1$ for all i) the identity

$$F^{(1)}_{n_0,m_0;k}(z,q) = \sum_{\substack{n_1,\ldots,n_k\geq 0\\ m_1,\ldots,m_k\geq 0}} \frac{z^{n_1}q^{\sum_{i=1}^{k}(n_i^2-n_im_i+m_i^2)}}{(q;q)_{m_0-n_1+m_1}}\begin{bmatrix} n_0\\ n_1\end{bmatrix}\begin{bmatrix} 2n_k\\ m_k\end{bmatrix}\prod_{i=1}^{k-1}\begin{bmatrix} n_i\\ n_{i+1}\end{bmatrix}\begin{bmatrix} n_i-n_{i+1}+m_{i+1}\\ m_i\end{bmatrix}. \tag{3}$$

Fix $k = 3$ and let $n_0, m_0 \to \infty$. Each coefficient of $z^n q^N$ on both sides stabilizes: on the right, $\begin{bmatrix} n_0 \\ n_1 \end{bmatrix} \to 1/(q;q)_{n_1}$ and $1/(q;q)_{m_0-n_1+m_1} \to 1/(q;q)_\infty$ coefficientwise; on the left the binomial chains telescope, $\begin{bmatrix} n_0 \\ n_1 \end{bmatrix}\begin{bmatrix} n_1 \\ n_2 \end{bmatrix}\begin{bmatrix} n_2 \\ n_3 \end{bmatrix} \to 1/((q;q)_{n_1-n_2}(q;q)_{n_2-n_3}(q;q)_{n_3})$ and likewise for the m-chain. (Stabilization: for a fixed monomial $z^n q^N$, all summands with any index exceeding $N + n$ contribute only above q^N, so both sides are finite sums of eventually constant coefficients.) Writing $(r, s) = (n, m)$, the left side becomes

$$\frac{\mathrm{FERM}_3^+(z,q)}{(q;q)_\infty} = T(z,q) := \sum_{\substack{r_1 \geq r_2 \geq r_3 \geq 0 \\ s_1 \geq s_2 \geq s_3 \geq 0}} z^{r_1} \frac{q^{\sum_{i=1}^3 (r_i^2 - r_i s_i + s_i^2)}}{(q;q)_{r_1-r_2}(q;q)_{r_2-r_3}(q;q)_{r_3}\,(q;q)_{s_1-s_2}(q;q)_{s_2-s_3}(q;q)_{s_3}\,(q;q)_{r_3+s_3}}, \tag{4}$$

where the right side of (3) has become $\mathrm{FERM}_3^+(z,q)/(q;q)_\infty$.

Step 2: Pochhammer split

Using $\dfrac{1}{(q;q)_{r_3+s_3}} = \dfrac{1-q^{r_3+s_3+1}}{(q;q)_{r_3+s_3+1}}$ and $q^{r_3+s_3+1} = q \cdot q^{r_3} \cdot q^{s_3}$ (a shift $(\rho_3, \sigma_3) \mapsto (\rho_3 + 1, \sigma_3 + 1)$ in (2)), term by term,

$$T(z,q) = S(e_3|e_3) - q\,S(e_2|e_2) = S_{13}\big((0,0,0)|(0,0,0)\big) - q\,S_{13}\big((0,0,1)|(0,0,1)\big). \tag{5}$$

Step 3: Uncu's modulo-13 theorem

Uncu (2024, Theorem `thm:m13`, first proved via Theorem `thm:n6m13`) proves the Kanade–Russell expression (`eq:mod13list`, last row) for the profile $(4, 3, 3)$:

$$H_{(4,3,3)}(z,q) = S_{13}\big((0,0,0)|(0,0,0)\big) - q\,S_{13}\big((0,0,1)|(0,0,1)\big). \tag{6}$$

Uncu's normalization $H_c = (zq;q)_\infty F_c/(q;q)_\infty$ (his eq. (`eq:HtoF`)) agrees with ours, and his F_c is the raw interlacing generating function.

Conclusion

Combining (4)–(6),

$$\frac{\mathrm{FERM}_3^+(z,q)}{(q;q)_\infty} = T(z,q) = H_{(4,3,3)}(z,q) = \frac{G_{(4,3,3)}(z,q)}{(q;q)_\infty},$$

hence $G_{(4,3,3)} = \mathrm{FERM}_3^+$. □

3 Numerical verification and remarks

Remark 4 (Verification artifacts)**.** *All steps were verified in exact arithmetic (`scratch/scripts/seed1_R2L4_d10` SageMath 10.9; log `scratch/tmp/seed1_L4_d10_chain_n12.log`):*

- *[FF] Warnaar's finite form* (3) *($a = +1$, $k = 3$) as an exact power-series identity (to q^{150}, computed at q^{300}) for all $0 \leq n_0, m_0 \leq 3$ — a transcription guard on the quoted proposition.*
- *[A] The split* (5) *at z-orders $n = 0, \ldots, 4$ to q^{150}.*

[B] $(q;q)_n(q;q)_\infty[z^n]\big(S(e_3|e_3)-qS(e_2|e_2)\big) = Q_{n,(4,3,3)}$ *against the reference engine (`seed8_R2L3_engine.sage` recursion: target-first kernel, exact* $\mathbb{Z}[q]$*, exact* $\Phi_3(q^m)$ *division,* Q_n *by Gauss inversion, Theorem Q/Corollary I of Seed 7 Layer 3) for* $n = 0,\dots,4$ *to* q^{150}.

[C] The finite fermionic sum of Corollary 2 equals the engine $Q_{n,(4,3,3)}$ exactly in $\mathbb{Z}[q]$ *for* $n = 0,\dots,12$ *(*$\deg Q_{12} = 1296$*), with all coefficients nonnegative; also* $Q_n(1) = 21^n$ *(*$K-1 = 21$ *at* $d = 10$*), as required.*

[D] The coefficientwise limit: $(q;q)_n(q;q)_\infty[z^n]T$ *equals the fermionic sum for* $n = 0,\dots,4$ *to* q^{150}.

Remark 5 (On the Kanade–Russell relation and Uncu's R_3 typo)**.** *The Layer-3 brief cautioned that Uncu's displayed* R_3 *relation for* $m \equiv 1 \pmod 3$ *contains a typo (two identical terms of opposite sign; his eq. (`R31`)). This is confirmed — the displayed relation is vacuous as printed — but it is* irrelevant to this proof*: at modulus* 13 *the Pochhammer split* (5) *produces exactly the pair* $(e_3|e_3)$*,* $(e_2|e_2)$ *appearing in Uncu's proved formula* (6)*, so no contiguous relation is needed. Structurally: at* $m = 3k-1$ *(the* $d = 8$ *case) the* $a = -1$ *finite form produces the* $(e_3|e_3), (e_2|e_2)$ *pair while the Kanade–Russell table lists* $(e_3|e_2), (e_2|e_1)$*, requiring one* R_3*; at* $m = 3k+1$ *the Kanade–Russell pattern* $H_{(c_1,c_2,c_3)} = S(e_{c_2}|e_{c_3}) - qS(e_{c_2-1}|e_{c_3-1})$ *evaluated at* $c_2 = c_3 = k-1$ *gives exactly the split pair.*

Remark 6 (Extensions)**.** *(i) For general* k*, the identical chain reduces `Con_cylindric-b` (first identity) at modulus* $3k+1$ *to the Kanade–Russell/Uncu expression for* $H_{(k,k-1,k-1)}$*, proved to date only for* $m = 13$*. The* $k = 3$ *case (*$d = 7$*, modulus* 10*) remains the smallest balanced case not proved anywhere. (ii) With Seed 2's* $d = 8$ *result, both balanced-orbit templates (*$a = \pm 1$*) are now realized; the remaining* $d = 10$ *core orbits require the* R_1/R_2 *word search plus new finite-form seeds, exactly as at* $d = 8$ *(synthesis-layer3 §4(v)).*

References

[1] S. Corteel, T. Welsh, *The* A_2 *Rogers–Ramanujan identities revisited*, Ann. Comb. 23 (2019).

[2] S. O. Warnaar, *The* A_2 *Andrews–Gordon identities and cylindric partitions*, Trans. Amer. Math. Soc. Ser. B (2023). (Local: `literature/tex/warnaar_A2_andrews_gordon/source.tex`; Proposition `Prop_finiteform` lines 2672–2687, proof lines 2784–2819; Conjecture `Con_cylindric-b` lines 767–785.)

[3] A. K. Uncu, *Proofs of modulo 11 and 13 cylindric Kanade–Russell conjectures for* A_2 *Rogers–Ramanujan type identities*, 2024. (Local: `literature/corteel-citations/tex/uncu_proofs_modulo11_13_cylindric_kanade_russell/main.tex`; S_{13} eq. (`eq:Sp1`) line 235; `eq:mod13list` lines 452–483; Theorem `thm:m13` line 573.)

[4] S. Kanade, M. C. Russell, *Completing the* A_2 *Andrews–Schilling–Warnaar identities*, 2022. (Contiguous relations; not needed in this proof.)